Lecture plan · for the lecturer

Indexing and query processing

Data engineering · University · Lecture 2

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Explain B-tree indexes.
- Reason about selectivity.
- Read query plans.

Activity: Tune one query

1. Warm-up (5 min): Ask students how many rows a lookup by student ID reads in a million-row table.
2. Explain (15 min): Pairs run EXPLAIN on a slow query and identify the scan.
3. Index (15 min): Add one index, rerun, and compare estimated and actual rows.
4. Measure (20 min): Time ten thousand inserts with and without the index.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- How many pages did the index read?
- When is a scan cheaper?
- What did writes cost?

Exit ticket

- Name one column in your system you would index and why.

Try it in your next lecture
Load a million-row table and let pairs run one slow query; they read the plan, add one index, rerun it, and then measure how much slower inserts became.

Seminar sheet · for students

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Name _____ Group _____ Date _____

1 **A yes/no column, evenly split. Will an index usually speed a filter on it?**

◯ Usually not ◯ Only for writes ◯ Yes, always

2 **Estimate the depth.**

With a fan-out of 100, a B+-tree over 10^6 keys needs _____ levels; at 10^8 keys it needs _____.

Word bank: 6 · 4 · 3 · 8

3 **An index on (dept, year) exists. Which queries can it answer with one contiguous range of entries? Choose all that apply.**

☐ WHERE dept = 'CS' AND year = 3 ☐ WHERE dept = 'CS' AND year > 2 ☐ WHERE year = 3

☐ WHERE dept = 'CS'

4 **Adding indexes to every column is a safe way to make an application faster.**

◯ True ◯ False

5 **Match each plan operation to when the optimiser prefers it.**

Sequential scan	• • • •	The index covers every needed column
Index scan	• • • •	Joining two large unsorted inputs on equality
Index-only scan	• • • •	The predicate matches most rows
Hash join	• • • •	A selective predicate on an indexed column

Answers · for the lecturer

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1 **Usually not**

Half the rows match, so a scan is cheaper.

2 **With a fan-out of 100, a B+-tree over 10^6 keys needs 3 levels; at 10^8 keys it needs 4.**

$100^3 = 10^6$ and $100^4 = 10^8$: a hundredfold growth adds one level.

3 **WHERE dept = 'CS'; WHERE dept = 'CS' AND year = 3; WHERE dept = 'CS' AND year > 2**

The index is sorted by dept, then year, so conditions on a prefix of its columns pick one contiguous range; rows with year = 3 are spread across every dept.

4 **False**

Each index slows writes and uses space; unselective indexes are ignored anyway.

5 **Sequential scan — The predicate matches most rows**
Index scan — A selective predicate on an indexed column
Index-only scan — The index covers every needed column
Hash join — Joining two large unsorted inputs on equality

The optimiser picks by estimated cost, not by how SQL is written.

For you to think about

Do your students ever see a table large enough for an index to matter? Where could you get one?